



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-2027
COURSE HANDOUT

Date: 05.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : MGTS U102
Course Title : INNOVATION AND DESIGN THINKING
Instructor-in-Charge : Leela Rani
Instructor(s)

Jyoti	Muthukumar G	Shriya Raina
Priya Christina Sande	Vinayak Devendra M	Pradipta Chattopadhyay
Sarita	Harshika Sondhi	

1. Course description:

Human-centered innovation, Design thinking (DT), consumer needs, creativity for new ideas/products (or re-design products) for solving problems, DT process' stages– inspiration, ideation & implementation, DT tools – stakeholder map, user persona, Hook's canvas, empathy map, AEIOU model, PoV, problem statement, stakeholder wish list, SCAMPER, prototyping & testing, MVP or minimum viable product, business model/canvas & business ecosystem. The pedagogical focus will be on hands-on approach to learning.

2. Scope and Objective of the Course:

The objective is to develop knowhow and skills that promote human-centered approach to innovation through design thinking (DT). It will aim to develop understanding of consumer needs & using creativity for new ideas/products (or re-design products) for solving problems. Course will also help understand prototyping, testing of new ideas & see the context of the larger ecosystem. Focus will be on understanding design thinking concept and tools for effective innovations that can solve consumers' needs holistically. Design Thinking process is laid across 3 stages- inspiration, ideation and implementation for new product development or re-design & concepts like MVP or minimum viable product, business model/canvas & business ecosystem will be also covered. The pedagogical focus will be on hands-on approach to learning.

3. Text Book:

- Design Thinking – A Beginner's Perspective, Author: E Balagurusamy. ISBN 13, 9789355329578. Publisher, McGraw Hill. Edition – 1st.
- From HBR's 10 Must Reads on Design Thinking:
 - Brown, T. (2008). Design thinking. *Harvard business review*, 86(6), 84.
Access from: <https://designthinkingmeite.web.unc.edu/wp-content/uploads/sites/22337/2020/02/Tim-Brown-Design-Thinking.pdf>
 - HBR article: 'Why Design Thinking' by J M Liedtka



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Access from: <https://hbr.org/2018/09/why-design-thinking-works>

Additional supportive material & notes will be shared on Nalanda from time to time.

4. Reference Books:

1. HBR's 10 Must Reads on Design Thinking (with featured article "Design Thinking" By Tim Brown)*
2. Design Thinking: A Comprehensive Textbook by Shalini Rahul Tiwari, Rohit Rajendra Swarup; ISBN: 9789357468640, 340 pages, Published by Wiley India, Publication Year: 2024
3. Michael Lewrick, Patrick Link, Larry Leifer, The Design Thinking Playbook: Mindful Digital Transformation of Teams, Products, Services, Business and Ecosystems, Wiley, May 2018 edition *
4. Isabell Osann, Lena Mayer & Inga Wiele, The Design Thinking Quick Start Guide: A 6-Step Process for Generating and Implementing Creative Solutions, Wiley, January 2020 edition.
5. Milind Haridas Ideas on Demand - A Crash Course on Creativity, Penguin Random House , India, 2023.
6. Jagadeesha M and others, Innovation and Design Thinking - Innovate, Integrate & Implement, Astitva Prakashan, 2024
7. Porus Munshi, Making Breakthrough Innovation Happen,, Collins Business, HarperCollins Publishers,2009
8. Harvard Business School Online Article Michael Boyles, Innovation in Business - What it is and Why it is Important.
Access from : <https://online.hbs.edu/blog/post/importance-of-innovation-in-business>
9. Patrick Healy (2016), 8 Types of Business Model and Value They Deliver
Access from : <https://online.hbs.edu/blog/post/types-of-business-models>
10. Strategyzer, What is a Business Model.
<https://www.strategyzer.com/library/what-is-a-business-model>
11. Don Norman: Human-Centered Design or Humanity-Centered Design?
<https://www.youtube.com/watch?v=eeE2Kl0LUWI>



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2. Module-wise Course Plan:

Sl.	Lecture topics	Ref. chap.	Mode: L/T*	Learning outcomes
1	Basic innovation concepts & HBR article - Brown, T. (2008). Design thinking	T1 Ch. 1 & T2 (A)	Lecture 1	Understand basics of innovation, creativity key concepts & rationale for design thinking, design thinker's personality.
2	HBR article 'Why Design Thinking' by J M Liedtka, examples of innovations around us	T2 (B) + notes	Lecture 2	Knowing how design thinking helps overcome problem of effective implementation of creativity solutions by an immersive customer experience, eg. of effective innovation from commercial and social sectors.
3	Introduction, groups, item & problem statement	Based on sl. 1 & 2	Tutorial 1	Introduction for all tutorials, group formation, item selection & problem statement construction for selected item
4, 5	Inspiration stage: Part 1	T1 Ch. 2 & select pages + notes	Lecture 3,4	Overview of concepts, tools for Inspiration stage of Design Thinking – knowing the consumer's context & opportunities, tools like user persona, empathy map, Hook's canvas, AEIOIU and stakeholder wish-list, business ecosystem and business canvas.
6,7	Inspiration stage: Part 2	Based on sl. 1 to 5	Tutorials 2 & 3	Hands-on application of tool(s) in group work => User profile & Hook's canvas
8	Ideation stage: Part 1	T1 : select pages + notes	Lecture 5	Developing problem statement, understanding consumer's ecosystem, generating ideas using DT tools - SCAMPER & evaluation matrix for feasibility, economic value and sustainability.
9	Ideation stage: Part 2	Based on sl. 5 & 8	Tutorial 4	Hands-on application of tool in group work => SCAMPER.
10	Implementation stage: Part 1	T1: Select pages + notes	Lecture 6	Explore prototyping basics. ideas to mock-ups and hi-fi prototypes, testing ideas on the ground and using feedback for next iterations to development, testing tools.



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Sl.	Lecture topics	Ref. chap.	Mode: L/T*	Learning outcomes
11	Implementation stage: Part 2 – Brainstorming session on ‘Prototype’	Based on sl. 11	Tutorial 5	Hands-on application of tool(s) in group work => Brainstorming for prototyping for re-design of item
12 -14	Group Presentations for evaluation, attendance in all sessions evaluated	Based on sl. 11 & 12	Tutorials 6, 7 & 8	Each group makes presentations based on their work for evaluation, attendance in all sessions evaluated

* Late entry into tutorial classes will be penalized by negative marks.

* All material & instructions will be shared on Nalanda – students must check regularly for lecture and tutorial updates.

4.Evaluation scheme	Duration	Weightage (%)	Date & Time	Nature of test (CB/ OB)
Component (ref Tut no)				
Tut 1 #	50 min	5#		---
Tut 2: Poster 1	50 min	20*		OB
Tut 3: Poster 2	50 min	20*		OB
Tut 4: Poster 3	50 min	20*		OB
Mid-Semester Test	40 min.	30	14/03	CB
Tut 5 # - Brainstorming session	50 min	5#		--
Tut 6, 7, 8: Group presentations	50 min	20* + 5,5#		--
Comprehensive Examination	NA	*best 3 out of 4		
# For attending 2 out of 4 (T1, T5 & {2 out of T 6,T 7, T 8 which is not your presentation session})				

Chamber and Consultation Hour: Will be announced by individual instructors on Nalanda

Notices: Notices concerning the course will be displayed on Nalanda.

Make-up Policy: Application for make-up exam only on urgent self-medical need (prior permission needed) – applications to be submitted only to I/C.

Note: Students are expected to know & follow all Institute rules.

Instructor-in-charge, MGTS U102